

Physics 1 (BAS 021) - Advanced Practice Exam (Model B)

Faculty of Engineering - BUC | Time: 3 Hours | Max Marks: 50

Question-1: Choose the correct answer (15 Marks)

1. The equation for the position of a train is $x=21at^2+bt^3$. The dimensions of the constant (b) must be:

a. $[L][T]^{-2}$ b. $[L][T]^{-3}$ c. $[M][L][T]^{-1}$ d. $[T]^{-3}$

2. A 2.0-mm diameter copper wire ($Y=11\times10^{10}N/m^2$) is stretched by 1.0%. The force required is approximately:

a. 2.1kN b. 3.5kN c. 6.9kN d. 5.4kN (تنبيه: القطر 2 مم يعني نصف القطر 1 مم، ويجب تحويله للمتر قبل حساب (πr^2) المساحة).

3. A satellite circles planet "Cruton" every 5.6 hours in an orbit of radius 12×10^6m . The magnitude of the gravitational force on a 50-kg satellite is:

a. 58 N b. 63 N c. 73 N d. 50 N

4. Two charges of $-15pC$ and $40pC$ are inside a cube with sides of $0.40m$. The net electric flux through the surface is:

a. $+1.1N\cdot m^2/C$ b. $+2.8N\cdot m^2/C$ c. $-1.1N\cdot m^2/C$ d. $-2.8N\cdot m^2/C$ (كولوم 10^{-12} هي pC ملاحظة: الـ).

5. If a worker pushes a crate at a constant speed, and then applies a force GREATER than the friction force, the crate will:

a. Move at a higher constant speed.

b. Accelerate (speed increases continuously).

c. Stop immediately.

d. Move at a constant velocity.

Question-2: Solve the following problems (35 Marks)

A. Dimensional Analysis & Power Law (7 Marks): The period (T) of a simple pendulum is believed to depend on the mass of the bob (m), the length of the string (l), and the acceleration due to gravity (g). Use **dimensional analysis** to find the power-law expression for the period. (Assume $T\propto m^x l^y g^z$).

B. Gravitation - The Falling Meteoroid (7 Marks): When a falling meteoroid is at a distance above the Earth's surface of **3.00 times the Earth's radius** (i.e., $h=3RE$), what is its **acceleration** due to Earth's gravitation? (Given: $G=6.67\times 10^{-11}$, $RE=6.37\times 10^6m$, $ME=5.98\times 10^{24}kg$). (فج: المسافة $r=RE+h=4RE$ هي).

C. Elasticity - The Shear Challenge (7 Marks): A child slides across a floor. The friction force on each foot is **20.0 N**. The footprint area is **14.0 cm²** and the sole thickness is **5.00 mm**. If the shear modulus of rubber is $3.00\times 10^6N/m^2$, find the **horizontal distance (Δx)** by which the sole is offset.

D. Electric Fields - The Vector Analysis (7 Marks): Two point charges are placed on the x-axis: $q_1=+7.0\mu C$ at $x=0$ and $q_2=-5.0\mu C$ at $x=0.3m$.

1. Calculate the **electric field components** (E_x, E_y) at point $P(0,0.4m)$.
2. Find the **magnitude and direction** of the total electric field at P .

E. Gauss' Law & Derivations (7 Marks):

1. State **Gauss's Law** in mathematical form.
2. Use Gauss's Law to **derive Coulomb's law** for the electric field of a point charge.